Journal of Hydrology 587 (2020) 124974

Contents lists available at ScienceDirect

Journal of Hydrology

journal homepage: www.elsevier.com/locate/jhydrol

Research papers

Evolutionary computational intelligence algorithm coupled with self-tuning predictive model for water quality index determination

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ARTICLE INFO

This manuscript was handled by Huaming Guo, Editor-in-Chief, with the assistance of Pedro J. J Depetris, Associate Editor.

Keywords:
Water quality index
Watershed management
Extreme Gradient Boosting
Genetic Programming
Extreme Learning Machine
Kinta River

ABSTRACT

Anthropogenic activities affect the water bodies and result in a drastic reduction of river water quality (WQ). The development of a reliable intelligent model for evaluating the suitability of water remains a challenging task facing hydro-environmental engineers. The current study is investigated the applicability of Extreme Gradient Boosting (XGB) and Genetic Programming (GP) in obtaining feature importance, and then abstracted input variables were imposed into the predictive model (the Extreme Learning Machine (ELM)) for the prediction of water quality index (WQI). The stand-alone modeling schema is compared with the proposed hybrid models where the optimum variables are supplied into the GP, XGB, linear regression (LR), stepwise linear regression (SWLR) and ELM models. The WQ data is obtained from the Department of Environment (DoE) (Malaysia), and results are evaluated in terms of determination coefficient (R^2) and root mean square error (RMSE). The results demonstrated that the hybrid GPELM and XGBELM models outperformed the standalone GP, XGB, and ELM models for the prediction of WQI at Kinta River basin. A comparison of the hybrid models showed that the predictive skill of GPELM (RMSE = 3.441 training and RMSE = 3.484 testing) over XGBELM improving the accuracy by decreasing the values of RMSE by 5% and 9% for training and testing, respectively with regards to XGBELM (RMSE = 3.606 training and RMSE = 3.816 testing). Although regressions are often proposed as reference models (LR and SWLR), when combined with computational intelligence, they still provide satisfactory results in this study. The proposed hybrid GPELM and XGBELM models have improved the prediction accuracy with minimum number of input variables and can therefore serve as reliable predictive tools for WQI at Kinta River basin.

1. Introduction

The continuous reduction in WQ remained a top global concern, especially in terms of industrial, agricultural and domestic utilization. Several factors including geological, weather, hydrological, industrial source (such as textile, laundry, and pharmaceutical etc.) and natural phenomena (physical processes) have adverse effect on the quality of water (Abba and Elkiran, 2017; Ke et al., 2015). In literature, water pollution is described as presence of detrimental substance(s) in water to the level that causes problems to living organisms. Therefore, monitoring water features to ensure the quality of is of paramount importance (Mahmoodabadi and Rezaei Arshad, 2018). WQ can be considered as set of chemical, physical and biological properties of water that can be utilized in forecasting the water quality (Sharma and Kansal, 2011). WQ helps in determining the chemicals concentration in the water. Thus, assessment of WQ can be describe as the analysis chemical, physical and biological properties of water. In the literature, several studies have been carried out in the area of water quality (Singh, 2017). Following this, WQI provide a single parameter describing the water quality reducing the huge number of parameters to a

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https://doi.org/10.1016/j.jhydrol.2020.124974
Received 10 March 2020; Received in revised form 11 April 2020; Accepted 14 April 2020
Available online 18 April 2020
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simpler expression, thereby making the monitoring and interpretation WQ effortless (Gazzaz et al., 2012). However, WQ is dependent on the ecosystem as well as human usages, such as industrial pollution, sewage, and wastewater; and both international and national agencies are committed to pollution control and WQ analysis (Bharti, 2011). Factually, there is no single variable can express the WQ efficiently, the WQ is normally determined by measuring multiple WQ parameters (Hameed et al., 2017). For this purpose, large amount of data is collected by the monitoring team which must be easily interpretable for decision-makers and the general public. In in this regard, numerous WQIs have been developed, which are defined based on WQ criteria and important parameters (Bharti, 2011).

WQI is a widely used measure in different parts of the world to solve problems of data management and to evaluate the successes as well as failures of management strategies for improving water quality. While numerous indices are used in summarizing WQ data to an understandable format, WQI is derived from numerous water characterization parameters to signifies water quality level (Abbasi, 2002). In essence, general approach to computing WQI involve passing numerous water quality parameters into computable functions and logical expressions that rates the wellbeing of a water body with a solitary number (Castilla-herná, 2014). The computed value is then rated from very bad to excellent based on existing rating scale which is understandable in ascertaining WQ by non-technical water managers, political decision-makers and the general public (Abbasi, 2002). Over the decades various WQI have been defined globally, these WQI help to represent the overall WQ in that particular area efficiently, including the United State National Sanitation Foundation WQI (NSFWQI) (Horton, 1965; Bharti, 2011), the Canadian Council of Ministers of the Environment WQI (CCMEWQI) (Sharma, 2002; Khan et al., 2004), the British Columbia Water Quality Index (BCWQI), and Oregon WQI (OWQI) (Debels et al., 2005). These indices are based on a comparison of the WQ parameters to regulatory standards and give a single value to the water quality of a source (Abba et al., 2019).

Despite the several paradigms that can be used to assess the quality of water, Malaysian Department of Environments (DoE) approved the use of a unified water quality indices in 1974, for analysis and ranking the level of contamination and pollution of Malaysian rivers. This recommended adaptation of WQI is called the DoE-WQI and served as method of choice for calculating the WQ index of local Malaysian rivers (Gazzaz et al., 2012; Yaseen et al., 2018a,b,c). DoE-WQI gives a standard WQ guidelines use to classify the WQ of Malaysian local rivers to five categories depending on their appropriateness for different usage including irrigation, domestic supplies and fish culture, water supply, recreational use, livestock drinking. The application used for computing the water quality in Malaysia is almost same as that employed in (Muhammad et al., 2015; Gazzaz et al., 2012; Yaseen et al., 2018a,b,c). However, the analysis and calculation of this method generally requires significant amounts of time and effort, which could lead to accidental errors during the sub-index computations; nevertheless, the method has proved to be highly effective and successful in practice based on scientific fact. For detailed manual calculation of the WQ sub-index for Malaysia, refer to (Hameed et al., 2017; Gazzaz et al., 2012).

Recent research shows that there is exponential increase in the use of Artificial Intelligence (AI), which represents an alternative, attractive, quick and direct computing tool for water quality modeling (Tiyasha et al., 2020; Sahoo & Patra, 2020; Gaya et al., 2020; Yasin and Karim, 2020; Karim and Kamsani, 2020). AI has the ability to minimize the error, effort and computation time (Bhagat et al., 2019). Artificial neural network (ANN), adaptive neuro-fuzzy inference system (ANFIS), and support vector regression (SVR) are among the popular artificial intelligence modeling methods developed for highly complicated process and handling data nonlinearity (Barzegar et al., 2016). However, several researchers have employed different combinations of AI-based models to predict the WQI (Wagener et al., 2019; Yaseen et al., 2018b). For example, Nourani et al. (2013) proposed the application of an ANN to monitor treated water quality. The results indicated that ANN has the potential to perform better than the conventional WQ method. Emamgholizadeh et al. (2014) applied ANN and ANFIS models simultaneously to predict the WQ variables of the Karoon River water, and the model results indicated the ANN model have a better predictive ability compared to ANFIS based models. In this regard, Barzegar et al. (2016) investigated the performance of standalone ANN and ANFIS, and hybrid wavelet-ANN and wavelet-ANFIS for the prediction of monthly averaged water salinity of Aji-Chay River in northwest Iran, where the results showed that ANFIS performance was superior to ANN. Hameed et al. (2017) examined two different models viz; Radial Basis Function NN (RBFNN) and Back Propagation NN (BPNN). The results from both models shows the reliable performance with improved accuracy.

Similarly, ANN model was employed to estimate the WQI at Langat River Basin, Malaysia, and the outcomes was compared with the traditional multilinear regression analysis. The obtained results demonstrated the effectiveness of the ANN model for the prediction of WQI (Juahir et al., 2004). Gazzaz et al. (2012) studied the potential of an ANN model to predict the WQI in Malaysia; the results demonstrated that the ANN model offered a reliable alternative to WQI computation and forecasting. Mohammadpour et al. (2014) studied and compared the potential of Support Vector Machine (SVM), BPNN and RBFNN techniques to predict the WQI in a wetland. For this purpose, different WQ variables at 17 monitoring points were evaluated. The results showed that SVM and FFBP outperformed RBF and therefore emerged as successful and reliable models for the prediction of WQI. The outcomes also indicated that the methods can reliably reduce time and computational burdens. More recently, Yaseen et al. (2018a,b,c) employed different types of AI-based models to predict the WQI. Abba et al. (2019) presented a performance comparison analysis of non-linear models based on RBFNN and Hammerstein-Weiner (HW) techniques and traditional linear modelling technique based on Generalized Linear Regression (GLR) and Autoregressive Integrated Moving Average (ARIMA) for Agra station of Yamuna river, India, and Kinta river in Malaysia. The simulation results established that the non-linear models outperformed the other models.

Based on the above literature, it is evident that studies in the field of WQ and WQI conducted around the world have shown the reliability of computational intelligence models (Tiyasha et al., 2020). Although there is a significant increase in the use of AI models in the field of WQI prediction, yet, many standalone AI models produce unsatisfactory results due to some related limitations to the identification of the appropriate input parameters or the selection of the internal models parameters. Particularly in applications involving very dynamic hydrological processes. Hence, to alleviate the aforementioned shortcomings, it is mandatory to develop an appropriate structure and to have a proper selection technique of the suitable variables. This has been proven to be significantly influential in the prediction ability for complex non-linear processes, in order to provide a reliable alternative to manual WQ computation. Hence, the objective of this study is to investigate the potential of XGB and GP in obtaining the feature importance, and then the abstracted inputs are used in the predictive model (i.e. ELM). For comparison purpose, the proposed models are compared with the stand-alone schema where the optimum combination of input variables are applied to the XGB, GP, ELM and Linear Regression (LR) models. The primary motivation behind the current study is to explore the abilities of the GP and XGB approaches for choosing the most dominant related variables. In this regard, hybrid models are established to increase the prediction accuracy of the non-tuned data intelligent models for mimicking the river WQ pattern. However, it is important to mention that since the development of this new algorithm, to the best of the authors' knowledge, no research or technical literature has been published in respect of application of XGB in WQI prediction and as an essential method of input selection. Because of its distinctive and outstanding features, this study, used XGB

algorithm in both input selection and WQI modelling. The remaining sections of the paper are arranged as follows: Section 2 presents the applied methodology for single model and proposed modeling schema adopted in the study, Section 3 described the case study and the data description, while the application of result and analysis are given in Section 4 and the paper ends with a conclusion in Section 5.

2. Applied methodology

2.1. Extreme gradient boosting (XGB)

The XGB algorithm has lately been dominating supervised machine learning applications, such as regression and classification (Pradhan and Sameen, 2020). It is an improvement of the gradient boosting technique introduced by Friedman (2001). It possesses the salient features of gradient tree boosting methods such as computational efficiency, high processing ability, and learning speed. It uses a more precise approximation to find the best decision tree model to achieve a higher speed and better performance. It is generally more than ten times faster than the conventional gradient boosting technique (Chen and Guestrin, 2016). XGB is an ensemble technique like Random Forests; this algorithm is attributed to the set of classification and regression trees (CART) as it differs from decision tree (DT), in that each of the leaves has a real score that helps in richer interpretation that beyond the classification which is the case in DT. In CART, a single tree is used which is incapable and weak to some extent, hence, an ensemble of multiple trees is proposed (Hadi et al., 2019; Xiong et al., 2007). However, although XGB has been implemented in a wide range of applications for example, stream flow (Hadi et al., 2019; Yu et al., 2020), shear strength (Zhang et al., 2020), and heat transfer (Qian et al., 2020). The use of the algorithm in WQI prediction in general and as a reliable means of input parameter selection with wavelet remain unexplored in the literature. For this purpose and given the excellent features of this algorithm, it is used in this work for both the modelling and input selection processes.

The ensemble of k trees, for *xi* (i.e. WQ variables) inputs dataset to forecast *yi* (i.e., WQI) target, can be expressed mathematically (see, Eq. (1)):

$$\hat{y} = \sum_{k=1}^{K} f_k(x_i), f_k \in F \tag{1}$$

where f_k is a function in a set of all likely function in CARTs, F the set of these functions, Optimization is the major objective of this function as presented in Eq. (2) (Chen et al., 2019) is:

$$obj(\theta) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{i=1}^{t} \Omega(f_i) \tag{2}$$

where l is the regularization function used to avoid overfitting, and Ω is the training loss function. A unary function is considered during the training, as it is difficult to train with all the functions simultaneously (f_i). Hence, representing the prediction value at step t as $\hat{y}_i^t$ leads to (Chen et al., 2019):

$$\hat{y}_i^t = \sum_{k=1}^{t} f_k(x_i) = \hat{y}_i^{t-1} + f_t(x_i) \tag{3}$$

Upon substituting the forecasted values given by Eq. (3), the tree that minimizes the objective function (Eq. (4)) is added to accomplish the performance of each iteration. Considering mean squared-error as the loss function, the cost function can be expressed as (Chen et al., 2015):

$$obj^t = \sum_{i=1}^{n} \left(y_i - (\hat{y}_i^{t-1} + f_t(x_i)) \right)^2 + \sum_{i=1}^{t} \Omega(f_i) \tag{4}$$

Which can also be in the form of Eq. (5) (Chen et al., 2015):

$$obj^t = \sum_{i=1}^{n} [2(\hat{y}_i^{t-1} - y_i) f_i(x_i) + f_t(x_i)^2 + \Omega(f_t) + constant \tag{5}$$

Mostly, a second-order Taylor's expansion is used due to the fact that logistic functions have a tendency of becoming complicated as expressed in Eq. (6) (Chen et al., 2015):

$$obj^t = \sum_{i=1}^{n} \left[l(y_i, \hat{y}_i^{t-1}) + g_i f_t(x_i) + \frac{1}{2} h_i f_t(x_i)^2 + \Omega(f_t) + constant \right. \tag{6}$$

where $g_i = \partial_{\hat{y}_i^{t-1}} l(y_i - \hat{y}_i^{t-1})$, and $h_i = \partial^2_{\hat{y}_i^{t-1}} l(y_i - \hat{y}_i^{t-1})$. The tree can be described by $f_t(x) = w_{q(x)}$, while the regularization function is expressed in Eq. (7).

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{7}$$

Therein T is the total number of trees, and q represents the scores on the leaves, that is used to assign the function of every data point to corresponding leaf. By eliminating the constants from Eq. (7), the objective function can be rewritten as in Eq. (8).

$$\begin{aligned} obj^t &\approx \sum_{i=1}^{n} \left[g_i w_{q(x_i)} + \frac{1}{2} h_i w_{q(x_i)}^2 \right] + \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \\ &= \sum_{j=1}^{T} \left[\left(\sum_{i \in I_i} g_i \right) w_j + \frac{1}{2} \left(\sum_{i \in I_i} h_i + \lambda \right) w_j^2 + \gamma T \right. \end{aligned} \tag{8}$$

where $I_i = \{i | q(x_i) = j\}$ refers to the jth leaf data index. Defining $G_j = \sum_{i \in I_i} g_i$ and $H_j = \sum_{i \in I_i} h_i$, one can write the objective function as in Eq. (9):

$$obj^t = \sum_{j=1}^{T} \left[G_j w_j + \frac{1}{2}(H_j + \lambda) w_j^2 + \gamma T \right. \tag{9}$$

Hence, for any tree structure $q(x)$, the performance can be obtained by using the best w_j and objective functions (see, Eq. (10) and (11), respectively).

$$w_j^* = -\frac{G_j}{H_j + \lambda} \tag{10}$$

$$obj* = -\frac{1}{2} \sum_{j=1}^{T} \frac{G_j}{H_j + \lambda} + \gamma T \tag{11}$$

Moreover, Eq. (12) is employed for optimization of the both the score gain and the tree in splitting a leaf into two.

$$Gain = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma \tag{12}$$

In Eq. (12) R and L indicate the score on new right and left leaves on the original leaves, and the regularization terms. If a branch gain is less, such branch is dropped from the tree. This algorithm is employed in obtaining the feature importance.

2.2. Extreme learning machine (ELM)

The ELM is a novel FFNN with a single hidden layer. It was first introduced and named by Huang et al. (2006). ELM was invented to eliminate the over-fitting and sluggish training speed issues attributed to the classical FFNN algorithms. In this method, a special least-square solution based on the Moore-Penrose inverse function is used to obtain the weights between the output layer and the hidden layer, while the bias of the hidden neurons and the weight between the hidden layer and the input layer are randomly initialized (Yaseen et al., 2016). The randomly initialized hidden neurons remain fixed. ELM can efficiently achieve a globally optimal solution, because of its universal

approximation abilities (Huang et al., 2006; Huang et al., 2015). In recent years, several researchers have used the ELM method in many areas of hydrological modelling owing to its robustness and high precision in streamflow prediction as in (Yaseen et al., 2019).

The ELM model was developed using a training data set$\{(x_1, y_1), \cdots, (x_t, y_t)\}$. In this context, the input vector denoted as $x_1, x_2, \cdots, x_t$ are defined as the WQ variables and the output vector $y_1, y_2, \cdots, y_t$ represents the observed WQI.

Mathematically, for N training sets ($t = 1, 2, \cdots, N$) where $x_t \in \mathbb{R}^d$ and $y_t \in \mathbb{R}$, an SLFN with H hidden nodes is given by (Huang et al., 2006) as in Eq. (13):

$$\sum_{i=1}^{H} B_i g_i(\alpha_i . x_t + \beta_i) = z_t, \tag{13}$$

where i is index of the hidden layer node, α_i and β_i, represent the weight and bias of the randomized layers, and d indicates the number of inputs. Similarly, $B \in \mathbb{R}^H$, $G(\alpha, \beta, x)$ and $Z(z_t \in \mathbb{R})$ are the predicted weights of the output layer, hidden layer neurons activation function, and model output respectively. The sigmoid function is found to be the best activation function for this study (Yaseen et al., 2020) (see, Eq. (14)).

$$G(x) = \frac{1}{1 + exp(-x)} \tag{14}$$

linear activation function is employed in output layer. As illustrated in Eq. (15), for ELM models, the error can be reduced to zero, by selecting the appropriate number of hidden neurons, randomized hidden layer biases (β) and input layer weights (α). Hence, the weights of output-layer may be found as (Huang et al., 2006):

$$\sum_{t=1}^{N} \|z_t - y_t\| = 0, \tag{15}$$

System of linear equations is used to find the value of B for any given input-output training samples is expressed in Eq. (16)

$$Y = GB \tag{16}$$

where G is expressed in Eq. (17):

$$G(\alpha, \beta, x) = \begin{bmatrix} g(x_1) \\ \vdots \\ g(x_N) \end{bmatrix} = \begin{bmatrix} g_1(\alpha_1 . x_1 + \beta_1) & \cdots & g_L(w_H . x_1 + \beta_H) \\ \vdots & \cdots & \vdots \\ g_1(\alpha_N . x_N + \beta_1) & \cdots & g_L(w_H . x_N + \beta_F) \end{bmatrix}_{N \times H} \tag{17}$$

and B is expressed in Eq. (18):

$$B = \begin{bmatrix} B_1^T \\ \vdots \\ B_H^T \end{bmatrix}_{H \times 1} \tag{18}$$

and Y is expressed in Eq. (19).

$$Y = \begin{bmatrix} y_1^T \\ \vdots \\ y_N^T \end{bmatrix}_{N \times 1} \tag{19}$$

where G denotes the output of the hidden layer. $\hat{B}$ can be estimated using Moore-Penrose inverse function + by inverting the hidden-layer matrix (see, Eq. (20)).

$$\hat{B} = G^{+}Y \tag{20}$$

Lastly, the estimated $\hat{y}$ which represent the predicted WQI for this study can be obtained as in Eq. (21).

$$\hat{y} = \sum_{i=1}^{H} \hat{B}_i g_i(\alpha_i . x_t + \beta_i) \tag{21}$$

2.3. Genetic programming (GP)

GP is one of the state-of-the-art machine learning algorithms that works on the basis of Darwinian's evolution theory (Igwe and Pillay, 2014). GP varies from traditional regression techniques, as it simply involves the determination of numerical coefficients for a function with a pre-specified structure (Olyaie et al., 2017). On the one hand, GP optimizes both the parameters and structure of an input-output map. Contrary to neural networks (NNs), user tuning is not required in developing GP-based models (Danandeh Mehr et al., 2013). In the GP technique, solutions are denoted as evolutionary parsing-trees, where the primary population is randomly generated. However, successive generations are obtained with the help of basic genetic operators; crossover, mutation and selection reproduction (Peirovi Minaee et al., 2019). GP has been effectively implemented to solve several problems in the fields of image processing, feedback control systems, electronic circuits, computer vision and data classification amongst others.

2.4. Linear regression (LR)

The traditional LR can be categorized into simple regression, which is used to estimate a single predictor and a single criterion variable, and multiple regression, which involves the estimations of more than one predictor but still with a single criterion variable. For this study, multiple regression analysis is used, which is also called multilinear regression analysis. According to (Chen and Liu, 2015), the multilinear regression model is the most widely used type of linear regression and involves an analysis whereby each value from the independent variables is related to a dependent variable value. The LR used in this study can be expressed as in Eq. (22):

$$y = b_0 + b_1 x_1 + b_2 x_2 + \ldots b_i x_i \tag{22}$$

where x_1, is the value of the ith predictor, b_0 is the regression constant, and b_i is the coefficient of the ith predictor.

2.5. Step-wise-linear regression (SWLR)

Regression analysis appeared to be the frequently used exploratory approach in citing dependencies between the two categories of variables (i.e. dependent and independent). While expressing dependencies between a dependent variable and multiple independent variables, it is paramount to find the optimum set of the variables that provide best prediction accuracy with respect to the dependent variable (Yasar et al., 2012). According to Çevik (2007) described the stepwise regression as a forward selection process which select the optimum set of independent variables through addition or deletion of the variable/variables with the greatest influence on the residual sum of squares. SWLR follows stepwise refinement of the variable(s) by rechecking the importance of previously added variables based on their partial sum of squares. When the partial sum of squares for any recently added variable(s) does not satisfy a base rule to remain in the models, the selection strategy is changed to deletion state and that variables are discarded gradually until every remaining variable meet the base rule (Yasar et al., 2012).

The concept of stepwise could be established using multiple linear regression as expressed in Eq. (22). All the independent variables x_i are entered into the regression equation one by one in accordance with their influence on the dependent variable Y. When each independent variable is entered, a series of t-tests are conducted for selected variables one by one in order to retain the statistically significant independent variables (Wu et al., 2020; Zhang and Chang, 2016). The process of entering or removing an independent variable from the regression equation (e.g., Eq. (22)) is a step of the stepwise regression. It also performs the significance test using the F-test at each step. If it is passed, the t-test results of each variable in the equation are checked until the P-value is less than 0.10. This process is repeated until no significant independent variables are entered into or removed from the

regression equation. From there, an optimized regression equation could be obtained (Lachniet and Patterson, 2006).

2.6. Proposed modeling schema

The main objective of the study is to find a tool for selecting inputs to improve the prediction of the WQI. Therefore, the schema of the study is to use GP and XGB as a nonlinear selection tool in addition to using them as the main model or prediction model. Afterwards, the selected inputs are to be entered into ELM and LR to assess their performance and compare them with the standalone models: GP, XGB, ELM, and LR. In detail, the methodology of the study is that the data was firstly normalized by using the following equation (see, Eq. (23)).

$$n_x = \frac{x - \min(X)}{\max(X) - \min(X)} \tag{23}$$

where nx is the normalized value, x is the original value, and X is all the values of the variable.

The normalized variables were entered directly into the used techniques (GP, XGB, ELM, LR, and SWLR) and this schema was called the standalone schema. Secondly, another schema is proposed based on the advantages of the GP and XGB models in obtaining the feature importance, and that schema is used to select only the most important features which contribute in the prediction of the WQI and impose them into ELM and LR. A visualization of the methodology is shown in Fig. 1. These two schemas are compared in order to obtain the advantage of selecting the inputs and the ability of GP and XGB in selecting the inputs.

Several parameters need to be considered in the modeling process in each of the used techniques. The GP model was run many times to find the best parameters and particularly the best Tree length and depth. The best mutation probability, generation, population size was found to be 15%, 1000, and 1000 respectively. After fixing these parameters, tree length and depth were also tuned by trial and error. For the XGB, grid tuning was used to obtain the best values of several parameters, such as maximum tree depth, Eta, and Gamma. In the ELM technique, 1 to 150 hidden neuron numbers were used, and the best is obtained by an algorithm that compares determination coefficient (R^2) and root mean square error (RMSE) of each one and the best performed hidden neuron was chosen. Three evaluation measurements (Metrics) were used, namely R^2 and RMSE, mean absolute percentage error (MAPE) (Ehteram et al., 2020; Khosravi et al., 2019) (see, Eqs. (24)–(26)).

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (WQI_{obs_i} - WQI_{pred_i})^2}{\sum_{i=1}^{N} (WQI_{obs_i} - \overline{WQI_{obs}})^2} \tag{24}$$

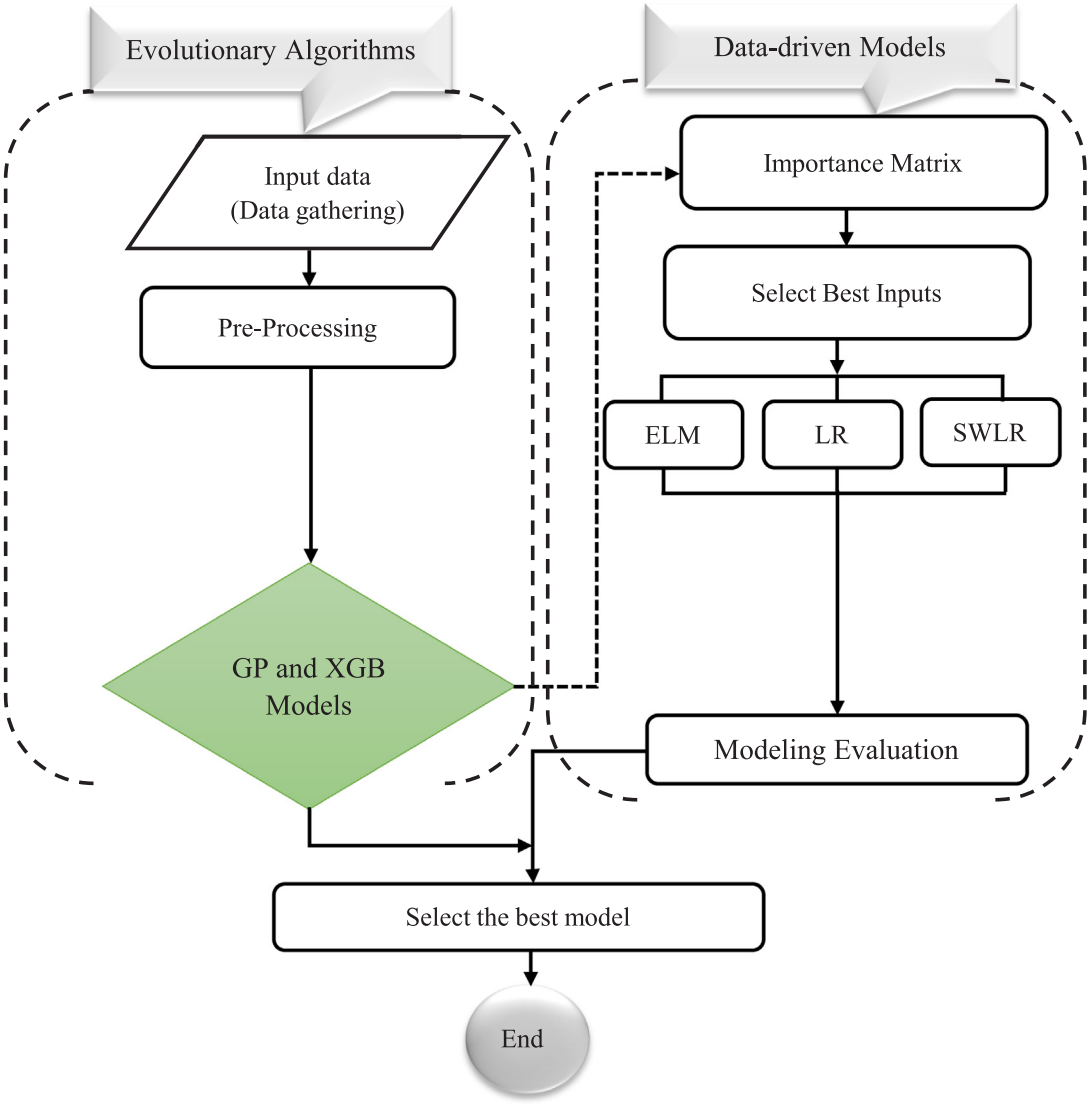


Fig. 1. The proposed modeling input selection schema for the WQI according to their importance.



Fig. 2. The location of the studied catchment.

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}(-WQI_{prd_i})^2}{N}} \quad (25)$$

$$MAPE = \frac{1}{N}\left[\sum_{i=1}^{N}\left|\frac{WQI_{obsi} - WQI_{comi}}{WQI_{obsi}}\right|\right] \quad (26)$$

where N, WQI, $\overline{WQI_{obsi}}$ and WQI_{predi} are data number, observed WQI, average value of the observed WQI and predicted WQI value, respectively.

3. Case study and data description

River Kinta span an area of about 2500 km^2 and a length of about 100 km within Kinta district. It supports farming activities, residential areas and industries (Fig. 2) across its three subdivisions (i.e. upstream, undulating and downstream). On the other hand, stations 2PK34 and 2PK25 are the center of the river, at the upstream there are stations 2PK22, 2PK24, and stations 2PK55 and 2PK19 are the downstream. Kinta river is largely utilized for forestry and recreation along the upstream. The densely populated Ipoh City is located at the center of the river, characterized by commercial and industrial activities. Kinta River dam constructed in the year 2000 with an attempt to raise the supply of Perak by around 25%, is the most important and second most important source of water to Ipoh and Perak respectively. It has a daily production capacity of 639,000 m^3 which is expected to be sufficient for Kinta Valley until 2020 (Ghani et al., 2007; Gazzaz et al., 2012). According to DoE Malaysia, 120 of the 189 rivers are monitored and controlled. Four decades ago, WQI was the recommended water pollution assessment and ranking index by the DoE along the rivers based on an approach named "Opinion Poll WQI (OP-WQI)" for calculating the water quality index of the local rivers. DoE estimates showed that 44.5% are clean rivers, 48.4% are slightly polluted, and 7.1% are polluted. Nonetheless, rainfall as the main source of water contribute 96% water supply to all kinds of surface water bodies (i.e., channels, rivers, impoundment, and reservoirs) (Ghani et al., 2007; Gazzaz et al., 2012).

The measured water quality data was obtained from DoE Malaysia for 12-years period between (2002 and 2013) with 301 instances. The data was recorded from several monitoring stations located on the Kinta River, Malaysia. The WQ parameters are SS, DO, BOD, COD, temperature (Temp), total solids (TS), PH and WQI. From the complete data 75% is used for training while 25% is set for testing the models. The statistical description of the data is given in Table 1. Despite there are other important WQ parameters, but the available parameters selected for the purposed of this research were considered as the one of the most dominants parameters in the determination of WQI as confirmed by other several studies such as (Muhammad et al., 2015; Gazzaz et al., 2012; Yaseen et al., 2018a,b,c). In addition, selecting these parameters was based on the non-linear input variables selection approach which is according the most sensitive to the target variables. It should note that another feasible alternative could also be used with regards to selection approach. However, the proposed modeling schema could also be applicable in modeling other important parameters.

From Table 1 the terms X_{min}, X_{max}, X_m, X_{mean}, S_x, and C_{sx} denote minimum, maximum, median, mean, standard deviation and skewness. The WQ parameters show variability trends in terms of the maximum and minimum values. The WQI from 43 to 96 mg/L, DO concentration varies from 0.28 to 9.93 mg/L. High values of WQ parameters indicate that the Kinta River receives significant non-biodegradable organic matter, which is due to the excessive anthropogenic activities in the recent years that significantly contribute to water pollution. Zhu and Heddam, (2019) characterized the WQ of Huaxi and Yipin River as being polluted with DO: 2.9–10 and DO 5.2–10.5, respectively. According to Elkiran et al. (2019) partially treated and untreated domestic and industrial wastewaters discharged into the river could lead to low DO concentration. According to Danandeh Mehr et al. (2013) and Olyaie et al. (2017), a very high C_{sx} can affect the performance of AI (ANN) considerably, while a low C_{sx} is more appropriate for modelling. The variation pattern of the WQ parameters confirms the non-linear

Table 1
Basic descriptive statistics of WQ variables.

Variables	X_{min}	X_{max}	X_m	X_{mean}	S_x	Csx
DO (mg/L)	0.28	9.93	4.4	4.82	2.12	0.29
BOD (mg/L)	1	54	3	4.09	4.70	6.97
COD (mg/L)	3	138	24	26.34	15.84	2.37
SS (mg/L)	0.5	2060	56	129.78	223.40	5.0
pH	4.54	8.29	7.01	6.99	0.42	−0.87
TEMP (°C)	23.06	35.49	27.96	27.94	1.99	0.05
TS (mg/L)	17	7613	134	216.67	481.62	12.59
WQI (mg/L)	43	96	74.7	74.32	11.24	−0.34

and complicated process of modeling DO concentration. Both mean and median indicated where the centre of the data is located, it can be seen in Table 1 that the values of mean are slightly larger than the median this indicated the positive direction of skewness with regards to mean values. According Litosseliti, (2010) the mean is best for un skewed data with no outliers while the median can tolerate skew and outlier's data.

Similarly, the validation of the datasets was conducted using k-fold cross-validation. The main aim of validation is to fit the model to the given dataset to obtain a reliable and accurate prediction on the datasets. For this purpose. In the k-fold cross-validation, the training and the validation datasets are independent; this advantage makes this method promising with a sound foundation for optimizing the model. The k-fold cross-validation approach has been widely implemented in various hydro-environmental studies even though other feasible alternatives may also be used (Tsioptsias et al., 2016).

4. Application results and analysis

The accuracy of intelligent models is affected by the use of numerous inputs. Several inputs selection methods have been reported in hydro-environmental literatures; popular among them includes; correlation, auto-correlation and principal components analysis. Nevertheless, these methods are frequently used for input/output linear relationships (Hadi and Tombul, 2018). This study employed two novel nonlinear input variables selection tools (GP and XGB) for choosing the most dominant related inputs to the targeted variable to improve the accuracy of the models' prediction. After that, the appropriately selected inputs are entered into non-tuned data intelligence and regression models to assess their performance in comparison with the standalone models; GP, XGB, ELM, LR and SWLR. The period used is sufficient to ensure the profile information of the Kinta River. Fig. 3 shows further analysis to indicate the direction and strength of association between the various stations via correlation matrix. For any computational intelligence techniques, it is essential to assess the correlation coefficient between the WQ variables. The correlation coefficient indicated the association between the selected stations and illustrate the proportionality relations between the dependent and the independent variables.

4.1. Results of GP and GPELM

This section demonstrates the results of running the GP standalone and hybrid ELM (GPELM) model. Table 2 only shows several important trials to save some space. According to Table 2, the best performing trial was trial 2 with a tree length of 20 and a tree depth of 100, which is also very close to trial 3 in terms of performance. With regard to complexity, trial 3 is the best as three input variables were excluded from the model, while in trial 2, only two were excluded and that reduces the complexity, and due to the difference in the performance between the two trials, trial 3 was chosen as the overall best.

The relevance in terms of importance of the features for the trials is also extracted and not only for the best model but in order to get the full image of the importance and prove the real importance of these features. Table 3 shows the obtained results. According to Table 2, trial 3 with a combination of DO, BOD, TS and COD has the highest performance accuracy in terms of model complexity. Regarding the performance accuracy of the trials (2 and 3), trial 2 outperformed trial 3 with slight increment in goodness-of-fit for both training and testing phases. Furthermore, the direct comparison of the trials clearly demonstrated that trial 2 to trial 6 achieved the best prediction skills and hence emerged as reliable combinations for the prediction of WQI at Kinta River. A goodness of fit criteria value equal 1 signifies a perfect estimation. But generally, a model with goodness of fit ≥ 0.7 is acceptable (Shukla et al., 2008). It is evident from Table 2 that R^2 values of trial 2–6 in this study >0.7.

The variables are sorted according to the mean of their relevance importance. After determining the importance of the inputs from the best model of the GP shown in the Table 4, only these variables are entered into the ELM in the same order of their importance. Table 4

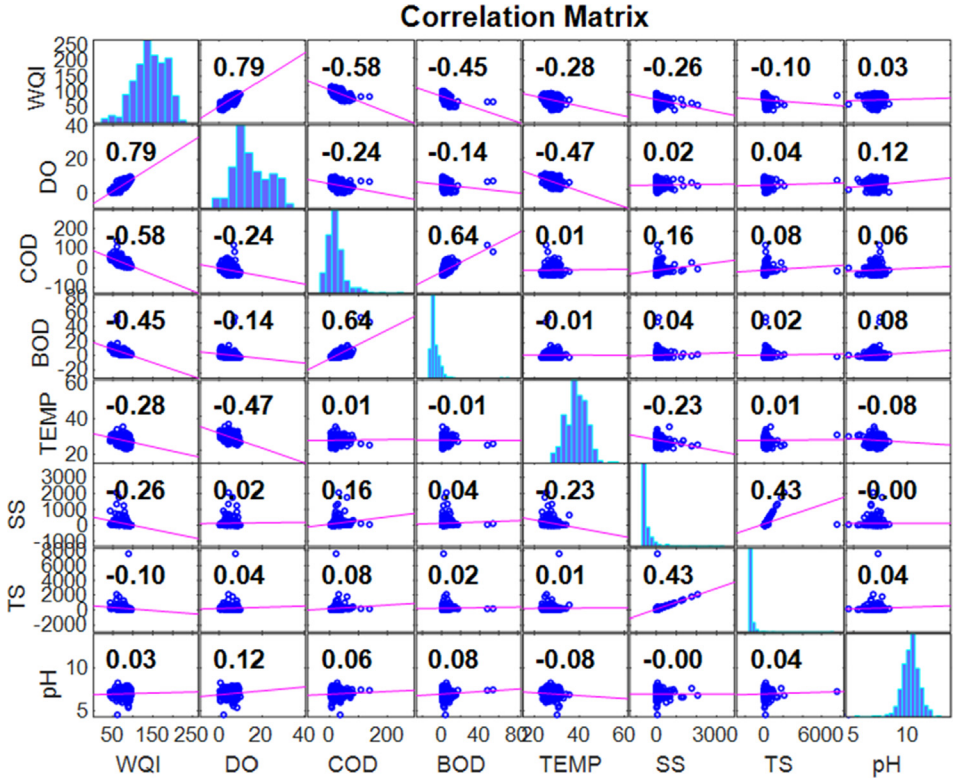


Fig. 3. Correlation matrix between the WQ variables.

Table 2
The trails of several GP models executions.

Trials	Best model depth	Best model length	R^2 (training)	R^2 (test)	RMSE (training)	RMSE (test)	Max depth	Max length	No. Used Var.
Trial 1	13	39	0.851	0.667	4.514	8.792	10	100	4
Trial 2	19	96	0.924	0.888	3.221	4.183	20	100	5
Trial 3	17	104	0.920	0.887	3.300	4.056	30	100	4
Trial 4	14	38	0.919	0.836	3.337	4.949	40	100	5
Trial 5	16	33	0.900	0.824	3.702	4.362	30	30	4
Trial 6	14	53	0.886	0.849	3.955	4.446	30	50	4
Trial 7	23	82	0.911	0.776	3.487	5.203	30	80	5
Trial 8	25	68	0.874	0.787	4.150	5.404	30	120	5

Table 3
The relevance importance of the associated input parameters.

Variable	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6	Trial 7	Trial 8	Mean
DO	0.147	0.142	0.181	0.15	0.154	0.187	0.158	0.145	0.158
BOD	0.145	0.154	0.145	0.145	0.154	0.146	0.145	0.145	0.147
TS	0.139	0.145	0.144	0.146	0.137	0.131	0.146	0.143	0.141
COD	0.144	0.135	0.132	0.143	0.144	0.136	0.145	0.143	0.14
pH	0.141	0.163	0.132	0.136	0.135	0.131	0.137	0.143	0.14
TEMP	0.142	0.133	0.134	0.141	0.137	0.134	0.134	0.138	0.137
SS	0.142	0.129	0.131	0.139	0.138	0.136	0.136	0.142	0.137

Table 4
The importance of the best model and the correlation with WQI.

Variable	GP Importance	XGB Importance	Cor. WQI
DO	0.614	0.384	0.795
TS	0.187	0.131	−0.446
BOD	0.117	0.236	−0.581
COD	0.092	0.168	−0.257
TEMP	0	0.064	0.034
pH	0	0.005	−0.278
SS	0	0.013	−0.098

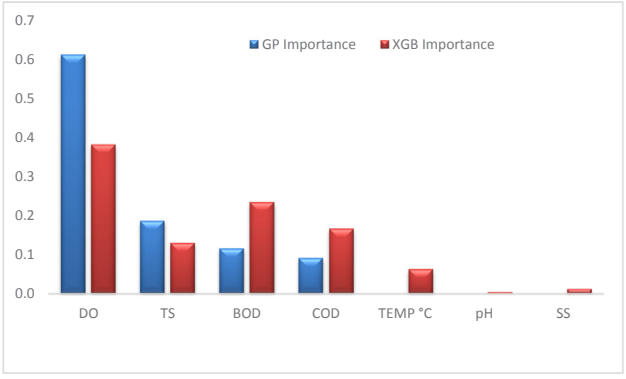


Fig. 4. The importance of the inputs of both GP and XGB schemas.

shows that the importance is not proportional to the correlation. For example, BOD has a higher correlation with WQI than TS, while the opposite is correct in terms of the importance as TS has higher importance than BOD. pH also has a higher correlation than COD; pH is not even entered in the prediction of the WQI as its importance is 0. This is due to the fact that the correlation is a linear index that only explains the linear relation between the dependent and independent variables, while in reality, the relation between them is not linear. The degree of importance of the inputs of both XGB and GP schemas is presented in Fig. 4.

It can be seen from Fig. 4 that Temp, pH and SS have zero importance according to the GP input selection approach; in other words, adding these variables has no or negligible impact on the prediction accuracy in the estimation of WQI. Although several studies, such as Ji et al. (2017), Elkiran (2018) and Shi et al. (2019) reported that Temp and pH are among the significant WQ variables, they are considered to be insignificant when using evolutionary input selection approach in this study. The results of the GPELM model in which only the selected variables are entered into the ELM model are shown in Table 5, and the best GP and ELM models results are also listed in order to compare them. According to Table 5, the GPELM has higher performance and the training and test subsets are very close, while there is a significant difference between them in the GP model and this could be due to the tree nature of the GP model which leads to overfitting.

A visual comparison between the three computational intelligence techniques shows the accurate performances in terms of RMSE and R^2. It can be vividly seen that integrating GP with the ELM model exhibits the highest performance in the prediction of WQI. Also, the numerical results indicated that for the forecasting of WQI at Kinta River, GPELM with R^2 = 0.908 and RMSE = 3.484 values in the testing phase was demonstrated to have excellent prediction skills in comparison to GP and ELM and therefore emerged as a reliable model. Fig. 5(a–c) present the scatter, and time series plots for the observed vs predicted WQI in the testing phase. It is clear from the figure that the best predictive model is GPELM, in which the prediction array was closer with the observed WQI. Another quantitative investigation of the results with reference to R^2 and RMSE in the testing phase depicted that the GPELM model increased by R^2 = 21% and R^2 = 56% and decreased by RMSE = 16% and RMSE = 9% compared to GP and ELM models during the testing stage, respectively.

4.2. Results of XGB and XGBELM

XGB was tuned for finding the best parameter by using grid tuning. Several parameters were tuned in which multiple models were created and to save space, only the important trials are listed in Table 6. XGB

Table 5
The performance of GP, ELM and GPELM.

Model	R^2 (training)	R^2 (test)	RMSE (training)	RMSE (test)	MAPE (training)	MAPE (test)
GP	0.920	0.887	3.300	4.056	3.8252	3.9523
ELM	0.834	0.852	4.765	3.816	5.0991	5.4136
GPELM	0.913	0.908	3.441	3.484	3.7901	4.018

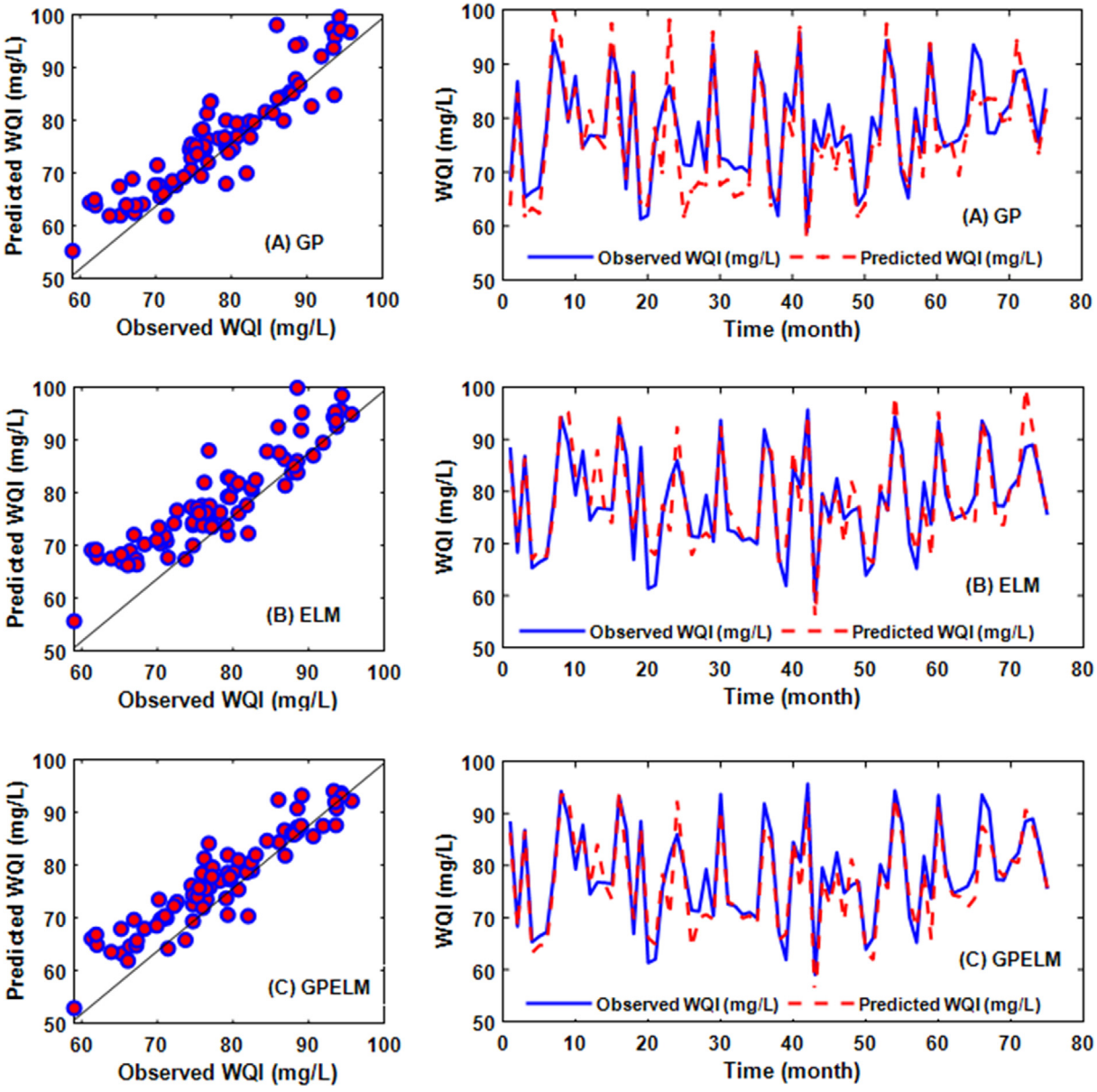


Fig. 5. Scatter, and time series plots for the observed vs predicted WQI (a) GP (b) ELM (c) GPELM in the testing phase.

hyper tuning techniques were implemented in order to achieve the optimum values of these parameters that lead to the best results.

From Table 6, trial 3 was chosen as the best trial with its parameters as it has the highest R^2 = 0.877 for the test subset and has the lowest RMSE = 4.062 for training and testing subsets. After choosing the best-tuned model, the importance matrix of the features (inputs or dependent variables) are also extracted, as listed in Table 4. The inputs after that are entered into the ELM according to their order in the importance matrix, as proposed by (Hadi et al., 2019). The results showed that the use of only the first four features in the importance matrix DO, BOD, COD, and TS obtains the best results and that is therefore identified as the best model. The results of the best XGB and best XGBELM model are shown in Table 7.

Although the XGB model has not received any attention in DO

Table 6
Several trials of the XGB tuning.

Trials	Max. Depth	Eta	Gamma	subsample	R^2 (training)	R^2 (test)	RMSE (training)	RMSE (test)
Trial 1	4	0.02	0.01	0.5	0.930	0.825	4.399	6.179
Trial 2	3	0.02	0.001	0.5	0.944	0.843	4.259	5.679
Trial 3	**1**	**0.1**	**0.01**	**0.5**	**0.933**	**0.877**	**3.036**	**4.062**
Trial 4	9	0.02	0.3	0.5	0.870	0.831	6.363	7.458
Trial 5	5	0.02	0.01	0.5	0.925	0.846	4.567	6.056
Trial 6	9	0.02	0.01	0.5	0.924	0.844	4.564	5.795
Trial 7	3	0.02	0.01	0.5	0.940	0.865	4.356	5.561
Trial 8	7	0.02	0.001	0.5	0.956	0.856	3.959	5.774

Table 7
The results of best XGB and Best XGBELM.

Model	R^2 (training)	R^2 (test)	RMSE (training)	RMSE (test)	MAPE (training)	MAPE (test)
XGB	0.933	0.877	3.036	4.062	3.3701	3.5123
ELM	0.834	0.852	4.765	3.841	5.0991	5.4136
XGBELM	0.905	0.910	3.606	3.816	4.0217	4.5182

prediction, this model has proved to be effective in streamflow prediction (Hadi et al., 2019). According to the presented results, XGB combinations (DO, BOD, COD, and TS) demonstrated the prediction skill with (R^2 = 0.933 and RMSE = 3.036 for training phase and R^2 = 0.877 and RMSE = 4.062 for testing phase). The predictability performance of XGB is therefore not surprising owing to it significant accuracy in Hadi et al. (2019). Fig. 6(a–c) show scatter plots between the observed and predicted values of the XGB, ELM and XGBELM. From the figure. It is clear that a closeness agreement between the observed and predicted values was attained. The overall comparison of the computational intelligence techniques indicates the satisfactory performance accuracy of GP, ELM, XGBGP and XGBELM models in both training and testing phases, which can be justified by considering the correlation coefficient (CC > 0.7). According to Legates and McCabe (1999), Moriasi et al. (2007) and Zhu and Heddam (2019) CC values higher than 0.70 are considered acceptable; thus, the results of all four models are acceptable.

The numerical predictive accuracy of the models was also evaluated, and the outcomes showed that GPELM and XGBELM yielded the best performance accuracy compared to the other single models. Although GP and XGB slightly outperformed the two hybrid models in some statistical indices, the hybrid models are considered the most reliable based on the agreement between training and testing phases. Detailed investigation for predicting WQI at Kinta River reveals that the model that obtains the best accuracy in the training phase may not necessarily represent as the best model. In addition, the impact of various WQ variables on WQI is different. This leads to the conclusion that the models that benefit from both the training and testing phases, such as GPELM and XGBELM, are capable of capturing the highly

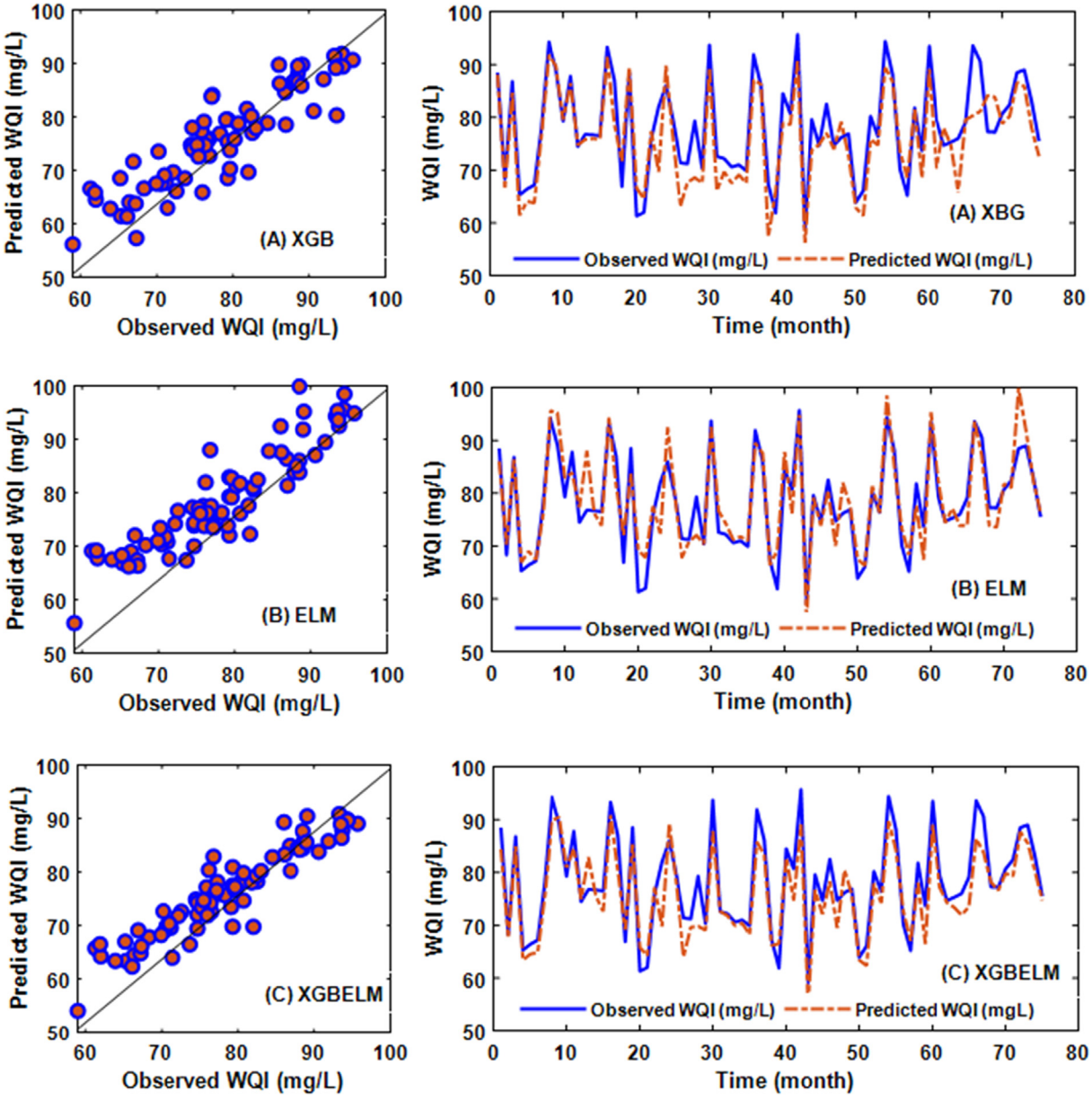


Fig. 6. Scatter and time series plots for the observed vs predicted WQI (a) XGB (b) ELM (c) XGBELM in the testing phase.

complex and nonlinear relationship. Generally, in hydro-environmental studies, there are several factors that affect the predictive accuracy of the models. However, potential factor affecting the accuracy in this study could certainly be the low correlation coefficient between WQI and pH, Temp and SS. Another reason could be due to the pollution increase caused by anthropogenic activities on the banks of the river, which drastically reduce the accuracy of the models. This statement is in line with the conclusion drawn by Zhu and Heddam (2019). The study conducted by Ji et al. (2017) also proved that there is a significant correlation between the DO WQ parameters except for Temp. The comparison of hybrid models depicted that the predictive skill of GPELM (RMSE = 3.441 and RMSE = 3.484) over XGBELM (RMSE = 3.606 and RMSE = 3.816), thus improving its accuracy by decreasing the values of RMSE by 5% and 9% for training and testing, respectively.

4.3. Results of linear regression

Linear regression was also used in this study as another model for comparison with the proposed models as a benchmark. The linear regression was firstly used with all variables as inputs; secondly, stepwise linear regression (SWLR) was used in order to obtain the best results by choosing only the important variables, and thirdly, the variables selected by the GP and XGB were entered to check the performance of the linear prediction of these variables.

The evaluation measurements of the three models are shown in Table 8. According to the results, the SWLR model has the highest performance in comparison with the other models. The model of GP or XGB with LR in which the inputs selected by GP and XGB is entered into the LR has the worst performance. The SWLR excluded pH, Temp, and TS as they have no significance in the prediction, as shown in Table 9. This exclusion increased the performance of the model in comparison with the model that used all the variables as inputs. In the model that used the inputs selected by GP/XGB in which SS, pH and Temp were excluded, the performance also deteriorated dramatically, although all the variables were significant according to Table 9. It is important to mention that the use of ELM alone has a very similar performance to the LR by including all the variables, but using them as the second step in the model after selecting the inputs ELM increased the performance dramatically, while for LR it dramatically decreased the performance. This is due to several reasons: the linear and non-linear nature of the GP and XGB in obtaining the importance of the features which is used in the selection of the inputs, the ELM non-linear prediction capability in comparison with the LR which is only linear and most of the non-linear relationships cannot be revealed.

Although regressions are often proposed as a reference model, when combined with computational intelligence they still provide satisfactory results. The performance of linear regression could be decreased for problems involving complex processes. As shown in Table 8, GP/XGBLR are inferior to the LR and SWLR in terms of predicting WQI values. Negligible changes were observed between LR and SWLR in both training and testing. Fig. 7(a–c) present the scatter and time series plots (LR, GP/XGBLR and SWLR) for the observed and predicted values in the testing phase.

Based on the visual examination of the predicted and observed WQI values during the testing stage, the prediction made by LR and SWLR are closer to the observed WQI than those made by GP/XGBLR. Further examination of the graph shows that with respect to goodness-of-fit (R^2 = 0.850) and statistical error (RMSE = 0.4074) in the testing phase, SWLR outperformed the other two models in term of prediction skills. Another pertinent conclusion can be made, from scatter plots obtained through LR, GP/XGBLR and SWLR in the testing phase denotes the ideal results in line 1:1. In particular, the closer the data points are to this line, the better the result produced by the model (Ji et al., 2017; Yaseen et al., 2019).

Table 8
The results of the three models of LR.

Model	R^2 (training)	R^2 (test)	RMSE (training)	RMSE (test)	MAPE (training)	MAPE (test)
LR all Inputs	0.837	0.848	4.724	4.096	5.2412	5.7103
SWLR	0.836	0.850	4.739	4.074	5.2449	5.8006
GP/XGBLR	0.786	0.829	5.412	4.434	5.9296	6.2183

Table 9
The significance of inputs n the prediction of WQI.

Variable	LR all inputs	SWLR	GP/XGBLR
Intercept	5.36E−07	2.85E−30	4.88E−25
DO	9.07E−35	2.80E−47	5.66E−40
BOD	3.75E−04	2.78E−04	5.27E−03
COD	5.12E−06	2.30E−06	2.32E−06
SS	1.03E−08	3.26E−11	–
pH	5.15E−01	–	–
Temp	4.76E−01	–	–
TS	8.96E−01	–	1.74E−02

The comparison of the predictive accuracies of all the models is illustrated by using a radar chart. Radar charts are essential in models' performance comparison because of its clarity and ability to show the tips of the variable scores in the dataset. In this study, radar charts are used in terms of CC for both the training and testing stages. Fig. 8 shows the radar chart illustrating the various CC values for all the models. It is observed from Fig. 8 that the CC values in the training and testing phases are (0.959, 0.941) for GP, (0.965, 0.936) for XGB, (0.955, 0.952) for GPELM, (0.951, 0.953) for XGBELM, (0.913, 0.923) for ELM, (0.914, 0.956) for LR (0.886, 0.941) for GP/XGBLR and (0.914, 0.956) for SWLR.

Moreover, box plots were used to demonstrate how close the comparisons of all the model's predictions are between the observed and predicted WQI, as shown in Fig. 9. The level of overall spread in the observed and forecasted values can be inferred from the various whiskers and quartiles. According to the graph, the spread of (GPELM and XGBELM) meticulously resembles the observed WQI. The exploratory analysis for all the models can be better pictured through the boxplots in Fig. 9. Boxplots are a powerful graphical representation of data that provide an overview and a statistical summary of a data set (Samsudin et al., 2019). According to Fig. 8, the closest of all the models to the observed values is selected to be the best model based on the mean value, the plot contained (box and whisker median, mean and staples). The extent of the spread values between the observed and predicted models indicates that the hybrid models (GPELM and XGBELM) emerged as the best among all the models.

As AI models are used, such models try to map the data by the training phase as much as they can but in the testing phase as these data are not shown to the model, they behave different and that is inherited property to the AI models. Also, for data-intelligence algorithms when the testing accuracy is greater than the training accuracy, there is a high chance that the model is overfitted (Fu et al., 2020). The model can be improved by reducing the bias and variance. Usually the best point is when both the bias and variance is low with the acceptable small gap between the training accuracy and testing accuracy. In another words, consider checking the class balance in training and testing data, unbalanced data may cause the model's test result to be biased towards one class. There are also some interesting points considering regarding regularization methods and dropout causing this behavior. But in case of Fig. 9a and b the box-plot trends give a good indication of how the values in the data are spread out they have the advantage of taking up less space, which is useful when comparing distributions between many groups or datasets (Samsudin et al., 2019). Hence the variation appeared based on the data spread but not with regards to quantitative

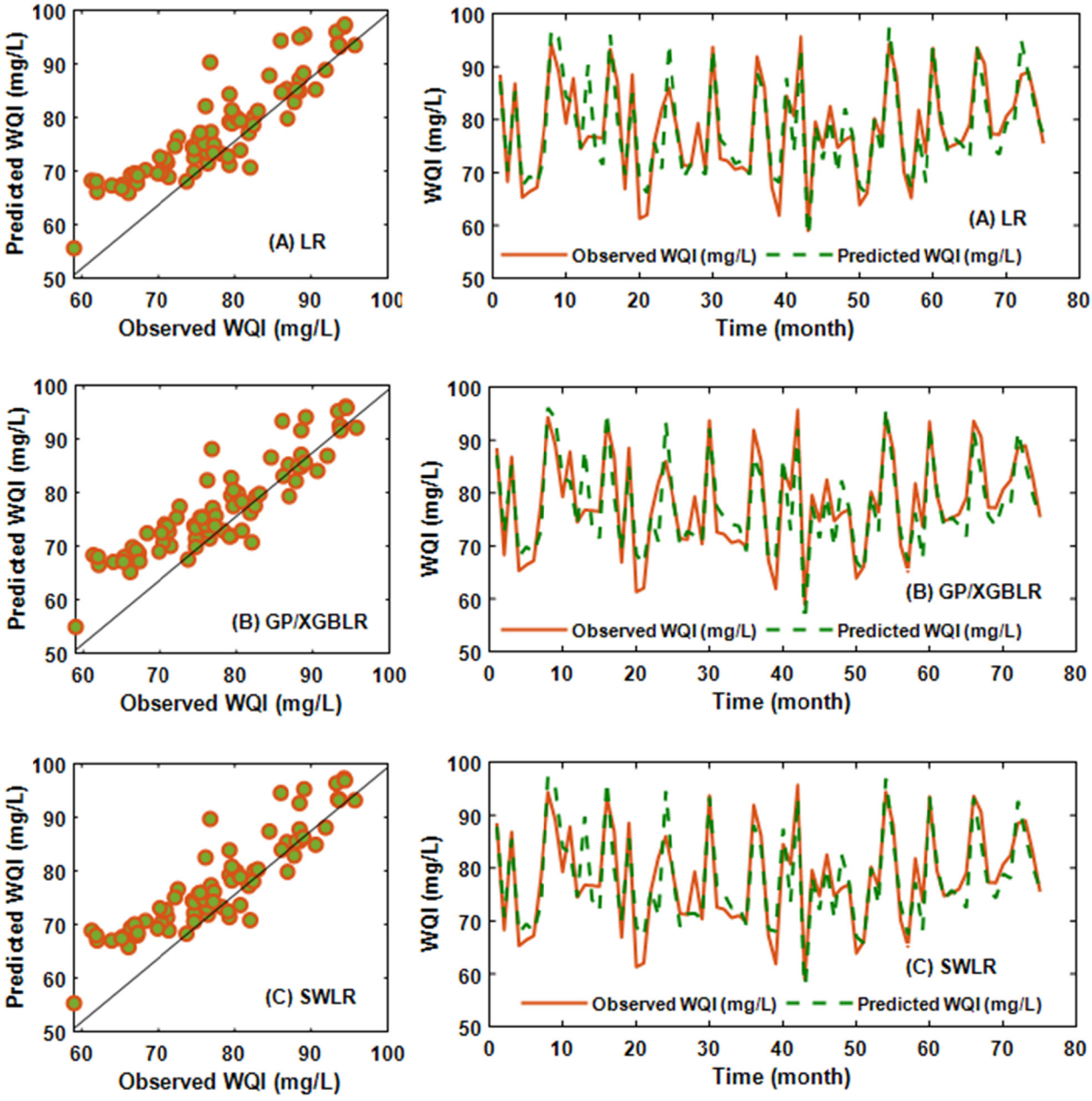


Fig. 7. Scatter and time series plots for the observed vs predicted WQI (a) LR (b) GP/XGBLR (c) SWLR in the testing phase.

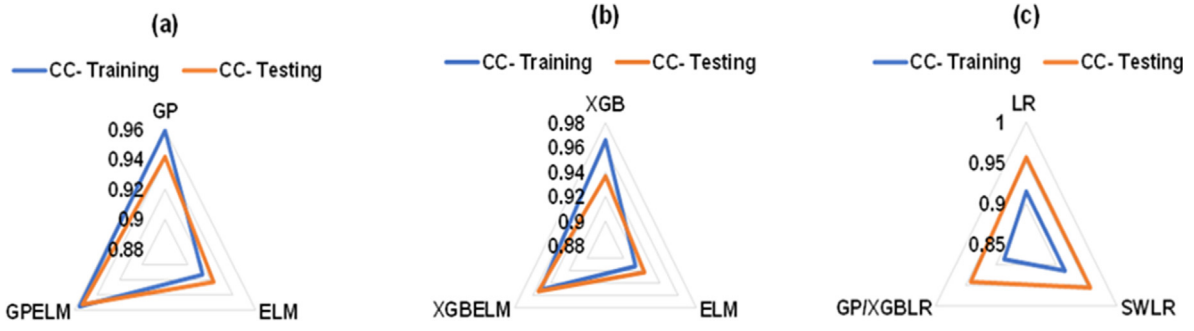


Fig. 8. Radar chart illustrating (a) CC for GP, ELM, and GPELM (b) CC for XGB, ELM, and XGBELM (c) CC for LR, SWLR, and GP/XBGLR.

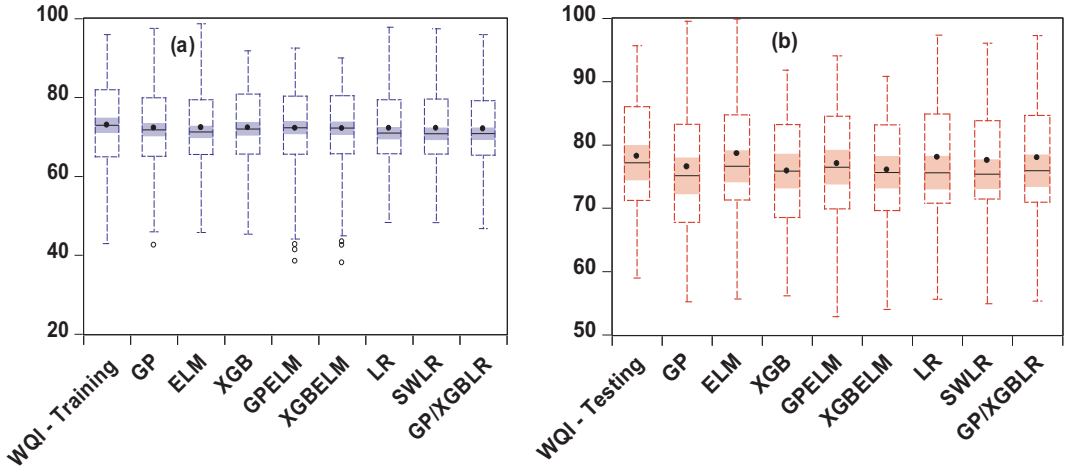


Fig. 9. Boxplot for observed and predicted models in (a) training phase (b) testing phase.

accuracy of the models.

5. Conclusion

The study was explored the abilities of two evolutionary nonlinear input variable selection techniques (GP and XGB) coupled with a non-tuned data intelligent model (ELM) for modeling the WQI in Kinta River basin, Malaysia. For this purpose, different trials were evaluated, and the best one was chosen as the best combination. The established modeling schema of the study was to use GP and XGB as a nonlinear selection tool in addition to using them as the main model or prediction models. Subsequently, the selected inputs are entered into ELM, SWLR and LR to assess the performance and compare them with the standalone models, namely GP, XGB, ELM, SWLR and LR. Following observations could be made from the comparison of the result:

- The use of the ELM model with all the variables exhibit identical performance with the LR.
- The capacity of the ELM model after selecting the important features by GP and XGB methods improves the performance of the prediction dramatically.
- The use of LR after selecting the inputs deteriorates the prediction performance.
- Stepwise LR increases the performance of the model predictability without selecting the same inputs selected by GP and XGB.
- GP and XGB methods can be successfully used as a selection tool in selecting the inputs especially in the existence of numerous features like the use of redundant information can deteriorate the performance.

The overall outcomes demonstrated that for modeling the WQI in Kinta River basin, Malaysia, the proposed computational intelligence techniques proved effective and satisfactory and hence can serve as a reliable prediction approach. Generally, the overall results of the study reveal the advantages of the GP and XGB, which can be successfully used as a selection tool in selecting the input variables in a highly nonlinear complex process. Furthermore, the results also suggest that other data intelligence models need to be adopted for further improvements in the prediction of WQI. Furthermore, using nature-inspired algorithm in the process of selecting suitable input variables provides a more informative attributes for the prediction matrix (Abdulwahab et al., 2019; Melin et al., 2013; Salih, 2019; Salih and Alsewari, 2019).

CRediT authorship contribution statement

S.I. Abba: Conceptualization, Validation, Investigation, Visualization. **Sinan Jasim Hadi:** Software, Validation, Investigation, Visualization. **Saad Sh. Sammen:** Conceptualization, Validation, Investigation, Visualization. **Sinan Q. Salih:** Validation, Investigation, Visualization, Validation, Investigation, Visualization. **R.A. Abdulkadir:** . **Quoc Bao Pham:** . **Zaher Mundher Yaseen:** Conceptualization, Validation, Investigation, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors acknowledge the historical water quality data availability by the Department of Environment (DoE) (Malaysia).

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